Swati Aggrawal

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PROFESSIONAL SUMMARY

MSc Computer Science candidate (AI/ML) with hands-on experience building AI infrastructure: document-parsing and semantic search systems, LLM-powered pipelines, and model evaluation frameworks. Proficient in Python, PyTorch, TensorFlow, and FastAPI, with deployed AI services on AWS. Practical experience in prompt engineering, RAG, embedding-based search, and benchmarking model performance for optimization. Comfortable working alongside engineering and product teams to identify AI-development challenges and ship efficient, usable solutions.

EDUCATION

Sapienza Università di Roma **May 2026**

Master of Science in Computer Science

- **Relevant Coursework:** Deep Learning and Applied AI, NLP, Computer Vision, Distributed Systems.
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning.
- Selected for a competitive international master's program.

Sharda University **July 2023**

Bachelor of Technology - CSE with AI and ML

- **Relevant Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort).

TECHNICAL SKILLS

- **Programming:** Python, SQL, C++, Bash
- **AI/ML Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn, FastAPI
- **AI Infrastructure & LLM Tooling:** LLM integration, Retrieval-Augmented Generation (RAG), Sentence Transformers, embedding models, semantic search, prompt engineering, document parsing/comparison
- **ML Systems:** Model training & evaluation pipelines, benchmarking, cross-validation, hyperparameter tuning, feature engineering
- **Cloud & Deployment:** AWS (ECS Fargate, CloudWatch), Docker, REST APIs, workflow automation, Git/GitHub
- **Data Analysis:** Pandas, NumPy, statistical analysis, exploratory data analysis, data visualization

PROJECTS

AI Semantic Text Compression using Transformer Embeddings

- Developed a semantic text compression pipeline using GTR-T5 embeddings, vector quantization, and text reconstruction to preserve semantic information.
- Designed and conducted systematic experiments across multiple quantization levels to evaluate the trade-off between compression efficiency and reconstruction quality.
- Built reproducible experimentation pipelines in Python for data preprocessing, embedding generation, quantization, and performance evaluation.
- Performed data cleaning, preprocessing, and experimental analysis using **Python, Pandas, and NumPy**.
- Evaluated model performance using **Cosine Similarity** and **ROUGE-L** metrics to benchmark semantic reconstruction quality.
- Achieved optimal performance at **6-bit quantization**, obtaining a cosine similarity of **0.964** and a ROUGE-L score of **0.776**.
- Conducted quantitative analysis to identify non-linear relationships between compression levels and semantic reconstruction accuracy.
- Documented experimental findings, model outputs, and performance trade-offs to support reproducibility and future optimization.
- Strengthened understanding of transformer-based embeddings, NLP pipelines, AI experimentation, and model evaluation through practical implementation.
- **Technologies:** Python, GTR-T5, Sentence Transformers, Pandas, NumPy,Git

DocDelta [Document Comparison API]

- Developed an NLP-powered document intelligence system using transformer embeddings and semantic similarity techniques for automated comparison of legal and contract documents.
- Designed algorithms to identify added, removed, and modified content while distinguishing numerical changes from textual modifications.
- Built an AI-powered document comparison pipeline that automated semantic analysis of document revisions, significantly reducing manual review effort.
- Leveraged Sentence Transformers and embedding-based retrieval techniques to analyze semantic similarity between document versions.
- Deployed the application using FastAPI, Docker, AWS ECS Fargate, and CloudWatch for scalable inference and monitoring.
- Evaluated inference performance and optimized deployment architecture, achieving approximately 840ms warm inference latency.
- Designed prompt engineering workflows to improve semantic document comparison using transformer-based language models.
- Evaluated LLM outputs and refined prompts to improve consistency and semantic accuracy.

RESEARCH EXPERIENCE

Continual Learning for Computer Vision (Master's Thesis) **Nov 25 - May 26**

- Designed and executed machine learning experiments to evaluate Vision Transformer performance across sequential learning tasks.
- Designed reproducible machine learning experimentation pipelines and comparative evaluation frameworks for transformer-based models.
- Conducted AI model benchmarking and comparative evaluation across multiple continual learning scenarios.
- Investigated catastrophic forgetting and knowledge retention in transformer-based visual learning systems.
- Automated experiment tracking and performance benchmarking across multiple continual learning scenarios.
- Conducted experiment tracking, performance analysis, and data processing using Pandas and NumPy.
- Documented experimental methodologies, findings, performance analyses, and future research opportunities to ensure reproducibility.
- **Technologies:** Python, PyTorch, Vision Transformers, Computer Vision

Machine Learning for Air Pollution [IEEE Publication] **May 2023**

- Published an IEEE conference paper reviewing machine learning approaches for air quality prediction.
- Conducted comparative statistical analysis of six machine learning models using multiple performance metrics across environmental datasets.
- Evaluated prediction-based and knowledge-discovery methodologies across six NAAQS pollutants.
- Identified real-time monitoring and IoT integration as important future directions for environmental analytics.